

Erica C.N. Robertson

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Education

- May 2022 – Current **Colorado State University**
PhD in Ecology through the Graduate Degree Program in Ecology, housed in the Department of Biology
- Aug 2016 – May 2020 **George Washington University**
Bachelor of Science in Molecular and Cellular Biology
- Jan – May 2019 **Universidad San Francisco de Quito**
IES exchange program at the Galapagos Academic Institute for the Arts and Sciences
- Aug 2015 – May 2016 **Parkland College**

Research Experience

- Aug 2022 – Present **PhD Student in the Kristen Ruegg Lab, Colorado State University**
- Investigating the genes underlying local adaptation in an alpine finch species, *Leucosticte tephrocotis dawsoni*
 - Integrating historical DNA to estimate changes in demography and genetic diversity over time in *Leucosticte australis*
 - Defining conservation units in *Riparia riparia* in collaboration with Alaska Department of Fish and Game
- Aug 2019 – July 2020 **Research Assistant in the Arnaud Martin Lab, The George Washington University**
- Designed and carried out experiments using CRISPR/Cas9 to perform gene knockouts
 - Reared butterflies and imaged wing patterns
 - Collected sequence data from NCBI and generated phylogenetic trees
- June – Aug 2018; Jan – May 2021 **Research Assistant in the Mark Davis Lab, University of Illinois**
- Performed qPCR analysis of environmental DNA (eDNA) samples targeting Odonata
 - Tested PCR primers for cicada microsatellite analysis
 - Completed and trained students in phenol-chloroform extractions of eDNA samples
 - Executed RADSeq digestions and adaptor ligation in preparation for sequencing
 - Trained and oversaw lab techs in various procedures such as PCR prep and gel electrophoresis
- June – Aug 2018 **Research Assistant in Chris Dietrich Lab, University of Illinois**
- Organized and cataloged hundreds of Membracoidae specimens
 - Performed dissections and Qiagen DNA extractions on Cicadellidae specimens in preparation for sequencing
- June – Aug 2018 **Research Assistant in Hugh Robertson Lab, University of Illinois**
- Manually annotated and analyzed Odorant Binding Protein genes in aphid genomes using the Apollo Genome Browser available at AphidBase

Publications

- Robertson, E. C. N., et al** (2026) The Genetic and Morphological Basis of Local Adaptation to Elevational Extremes in an Alpine Finch. *Ecology and Evolution*. <https://doi.org/10.1002/ece3.72962>
- Mastretta-Yanes, A., et al.** (2024) Multinational evaluation of genetic diversity indicators for the Kunming-Montreal Global Biodiversity Framework. *Ecology Letters*, 27, e14461. <https://doi.org/10.1111/ele.14461>

Hanly, J., **Robertson, E. C. N.**, Corning, O. B. W. H., Martin, A. (2021). Porcupine/Wntless-dependent trafficking of the conserved WntA ligand in butterflies. *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution*. <https://doi.org/10.1002/jez.b.23046>

Robertson, H. M., **Robertson, E. C.**, & Walden, K. K. (2019). The chemoreceptors and odorant binding proteins of the soybean and pea aphids. *Insect Biochemistry and Molecular Biology*, 105, 69-78. <https://doi.org/10.1016/j.ibmb.2019.01.005>

Pending Publications

Forester, B. R., **Robertson, E. C. N.**, DeSaix, M., et al. (2026) A widespread gap in the U.S. Endangered Species Act: Risk of genetic erosion within populations. *In Review*. <https://doi.org/10.32942/X2KX06>

Nikunj, Goel, **et al.** (2026) Identifying adaptive variation in spatially structured populations using low-coverage whole-genome sequencing data. *In Review*. <https://doi.org/10.64898/2026.01.19.700474>

Robertson, E. C. N., et al. (2026) The Importance of Evolutionary Scale in Diagnosing Population Decline in Bank Swallows. *In Prep*.

Ruegg, K. C., **et al.** (2026) Loss of Adaptive Capacity Drives Climate Vulnerability Across Taxonomic Scales in an Alpine Specialist Species Complex. *In Prep*. <https://doi.org/10.64898/2026.05.13.724772>

Presentations

Robertson, E. C. N., et al (2026) At the Right Scale: Using Evolutionary Significant Units to Diagnose Population Decline in Bank Swallows. *Oral Presentation, American Ornithological Society Annual Meeting, Amherst, MA*.

Robertson, E. C. N. (2026) The Bird Genoscape Project: A full annual cycle approach to avian ecology, evolution and conservation. *Oral Presentation, San Fernando Valley Audubon Society, Van Nuys, CA*.

Robertson, E. C. N. (2026) The Bird Genoscape Project: A full annual cycle approach to avian ecology, evolution and conservation. *Oral Presentation, Denver Audubon, Denver, CO*.

Robertson, E. C. N., et al (2026) At the Right Scale: Using Evolutionary Significant Units to Diagnose Population Decline in Bank Swallows. *Oral Presentation, Front Range Student Ecology Symposium, Colorado State University, CO*. (Outstanding Oral Presentation)

Robertson, E. C. N. (2025) Local adaptation in Brown-capped Rosy Finch. *Oral Presentation, Denver Audubon, Denver, CO*.

Robertson, E. C. N., et al (2024) Novel genes underlying locally adapted traits in an alpine finch species. *Oral Presentation, American Ornithological Society Annual Meeting, Estes Park, CO*.

Robertson, E. C. N. (2024). Assessing Local Adaptation and Underlying Genes in an Alpine Finch. *Oral Presentation, Front Range Student Ecology Symposium, Colorado State University, CO*. (Outstanding Oral Presentation)

Research Grants

2023 **Assessing Climate Risk in Alpine Finch Species**, Colorado Field Ornithologist (\$850)

2024 **Assessing Climate Risk in Alpine Finch Species**, Lois Webster Grant, Denver Audubon Society (\$3900)

2024 **Assessing Local Adaptation and Underlying Genes in an Alpine Finch Species**, GDPE Research Grant, Colorado State University (\$600)

Teaching and Course Design

2026 **ECOL592-The Science and Policy of Species Conservation**
Codesigned and assisted in running a graduate level course on the integration of science and policy in species listing decision making at the global, federal, and state levels

Spring 2022 – Spring 2026	Graduate Teaching Assistant, CSU GTA for LIFE102 (1 semester) and BZ350 (5 semesters)
2025	ECOL592-Nature Journaling Conceptualize, designed, and ran graduate level course on nature journaling
Spring 2025	Genetic Indicators Workshop Developed workshop materials for national workshop on genetic diversity indicators presented to member of the Namibia government and academic groups
Aug 2019 – May 2020	Head Learning Assistant, GWU

Work Experience

Jan 2024 – Current	Graduate Research Assistant, CSU
January – May 2025	Genomics Consultant for the University of Namibia
May 2017 – Dec 2019	Lab Prep Assistant, GWU Biology Department
Sep 2016 – Dec 2017	Explainer, Smithsonian National Air and Space Museum

Service and Outreach

2024 and 2026	Poster Presentation Judging For CSU's Celebrate Undergraduate Research and Creativity Showcase
2024 and 2025	Mentored Undergraduate Students As part of CSU's MURALS program and independently through the Ruegg Lab
2023	Teacher for Bird Camp An outreach program run by the Bird Genoscape Project

Honors and Awards

February 2026	Outstanding Oral Presentation The 2026 Front Range Student Ecology Symposium
Fall 2024-Spring 2025	Sharon E. Kabes Scholarship (\$1200) From Colorado State University
February 2024	Outstanding Oral Presentation The 2024 Front Range Student Ecology Symposium
Fall 2023-Spring 2024	Dr. Heather M. Rueth Ecology Scholarship (\$1900) From Colorado State University
Fall 2016, Spring 2017, Spring 2020	Dean's List Attaining a GPA above 3.75/4.0
May 2020	Received <i>cum laude</i> honors From George Washington University